

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	Phase 1 Open-label, Multicenter Study of MK-1454 Administered by Intratumoral Injection as Monotherapy and in Combination With Pembrolizumab for Patients With Advanced/Metastatic Solid Tumors or Lymphomas
Short Title:	Study of Ulevostinag (MK-1454) Alone or in Combination With Pembrolizumab (MK-3475) in Participants With Advanced/Metastatic Solid Tumors or Lymphomas (MK-1454-001)
Protocol Number:	1048756961226174809
NCT Number:	NCT03010176
Sponsor Name:	Merck
Investigational Product:	pembrolizumab
Phase:	PHASE1
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Part 1: Percentage of Participants Who Experienced a Dose-limiting Toxicity (DLT) Per Common Terminology Criteria for Adverse Events, Version 4.0 (CTCAE 4.0): DLTs were assessed during the first cycle (21 days) \& are defined as: Grade (Gr) 4 nonhematologic toxicity; Gr 4 hematologic toxicity lasting ≥7 days, except thrombocytopenia, Gr 4 thrombocytopenia, Gr 3 thrombocytopenia (if associated with clinically significant bleeding); nonhematologic adverse event (AE) ≥ Gr 3 (with exceptions); Gr 3 or 4 nonhematologic lab abnormality (if medical intervention is required, leads to hospitalization, or persists for >1 week); Gr 3 or 4 febrile neutropenia; drug-related toxicity that causes treatment discontinuation or dose delay >7 days between consecutive doses during Cycle 1; drug-related toxicity that causes a >2 week delay in Cycle 2 initiation; elevated aspartate aminotransferase or alanine aminotransferase lab value that is ≥3× upper limit of normal (ULN) \& an elevated total bilirubin value ≥2× ULN \& an alkaline phosphatase value <2× ULN, in which no alternative reasons can be found; ≥Gr 2 immune-mediated uveitis; or Gr 5 toxicity. 2. Parts 1 and 2: Number of Participants Who Experienced One or More Adverse Events (AEs): AEs are defined as any unfavourable and unintended sign (including an abnormal laboratory finding, for example), symptom, or disease temporally associated with the use of study treatment or protocol-specified procedure, whether or not considered related to the study treatment or protocol-specified procedure. Any worsening (i.e., any clinically significant adverse change in frequency and/or intensity) of a pre-existing condition that is temporally associated with the use of the study treatment, is also an AE. 3. Parts 1 and 2: Number of Participants Who Discontinued Study Drug
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Due to an AE: AEs are defined as any unfavourable and unintended sign (including an abnormal laboratory finding, for example), symptom, or disease temporally associated with the use of study treatment or protocol-specified procedure, whether or not considered related to the study treatment or protocol-specified procedure. Any worsening (i.e., any clinically significant adverse change in frequency and/or intensity) of a pre-existing condition that is temporally associated with the use of the study treatment, is also an AE.

Secondary Obj.:

1. Parts 1 and 2: Ulevostinag Area Under the Plasma Drug Concentration-Time Curve From Time Zero to 24 Hours (AUC₀₋₂₄) 2. Parts 1 and 2: Ulevostinag Minimum Plasma Concentration (C_{min}) 3. Parts 1 and 2: Ulevostinag Maximum Plasma Concentration (C_{max})

2.2 Background & Rationale

The purpose of this study is to identify a maximum tolerated dose (MTD) or maximum administered dose (MAD) of ulevostinag alone and of ulevostinag in combination with pembrolizumab in participants with advanced/metastatic solid tumors or lymphomas in Part 1, and to evaluate the safety and efficacy of ulevostinag via intratumoral (IT) injection in combination with pembrolizumab in selected solid tumors in Part 2.

3. Study Design & Methodology

Design Framework: Type: INTERVENTIONAL, Phase: PHASE1, Status: COMPLETED

Target N: 156

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

All Arms and Cohorts (Parts 1 and 2):

- * Has ?1 injectable lesion which is measurable and amenable to injection and biopsy.
- * Has an Eastern Cooperative Oncology Group (ECOG) Performance Status score of 0 or 1.
- * Demonstrates adequate organ function within 7 days prior to treatment initiation.
- * Female participants of childbearing potential must be using a contraceptive method that is highly effective or be abstinent from heterosexual intercourse (on a long-term and persistent basis) during the intervention period and for at least 130 days after the last dose of study intervention and agrees not to donate eggs to others or freeze/store for personal use for the purpose of reproduction during this period. Male participants must agree to refrain from donating sperm PLUS either be abstinent from heterosexual intercourse (on a long-term and persistent basis) OR agree to use a male condom plus partner use of an additional contraceptive method when having penile-vaginal intercourse contraception, unless confirmed to be azoospermic (vasectomized) during the intervention period and for at least 130 days after the last dose of study intervention.
- * Human Immunodeficiency (HIV)-infected participants must meet these additional criteria: a) Has laboratory-test-documented HIV-1 infection; b) Has well-controlled HIV on anti-retroviral therapy (ART), defined as: 1) must have a cluster of differentiation (CD4+) T-cell count ≥ 350 cells/mm³ at time of screening; 2) must have achieved

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and maintained virologic suppression defined as confirmed HIV ribonucleic acid (RNA) level below 50 or the lower limit of quantification (LLOQ) using the locally available assay at the time of screening and for ≥12 weeks prior to screening; and, 3) must have been on a stable regimen, without changes in drugs or dose modification, for ≥4 weeks prior to study entry (Day 1).

All Part 1 Arms:

-Has ≥1 distant, discrete non-injected lesion which is amenable to biopsy. This lesion must be measurable as defined by the response criteria used to assess the participant (RECIST 1.1 for solid tumors or revised International Working Group [IWG] criteria for lymphomas).

Part 1 Arm 1: Ulevostinag (Cut/Subcut Lesions) and Part 1 Arm 2: Ulevostinag+Pembro (Cut/Subcut Lesions):

- * Has a histologically- or cytologically-confirmed advanced/metastatic solid tumor or lymphoma by pathology report and who has received, or been intolerant to, all treatment known to confer clinical benefit. Solid tumors and lymphomas of any type are eligible for enrollment.

- * Has stage III or stage IV disease that is not surgically resectable. Stage IIB (T3N0M0B0-1) cutaneous T cell lymphoma (CTCL) participants are eligible.

Part 1 Arm 3: Ulevostinag+Pembro (Visceral Lesions):

- * Has stage III or stage IV disease that is not surgically resectable.

- * Has metastatic liver involvement that does not exceed one third of the total liver volume in participants to be treated by liver IT injection. Hepatocellular carcinoma participants are excluded from eligibility of IT liver injection.

All Part 2 Expansion Cohorts:

Part 2 Cohort A: HNSCC Anti-PD-1/PD-L1 Refractory:

- * Has HNSCC of the oral cavity, oropharynx, hypopharynx, or larynx; anti-PD-1/PD-L1 refractory metastatic or recurrent. Participants may not have a primary tumor site of the nasopharynx (any histology).

- * Has histologically confirmed Stage III, IVa, or IVb disease per TNM (Tumor, Nodes, Metastasis) staging, American Joint Committee on Cancer (AJCC, 8th edition), with recurrent or persistent disease after definitive chemoradiation, deemed unresectable and considered refractory to both platinum-based combination chemotherapy and anti-programmed cell death-ligand 1 (anti-PD-1/PD-L1) antibody therapy.

OR

- * Has histologically confirmed Stage IVc disease per TNM staging, AJCC 8th edition, considered refractory to platinum-based combination chemotherapy and anti-PD-1/PD-L1 antibody therapy.

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Part 2 Cohort B: Anti-PD-1/PD-L1 TrT-Naïve or Refractory TNBC:

- * Has confirmed unresectable locally advanced or metastatic TNBC as locally determined according to the American Society of Clinical Oncology-College of American Pathologists (ASCO-CAP) guidelines on a newly obtained core or excisional biopsy from a metastatic, not previously irradiated, tumor lesion.
- * Has received at least one prior systemic treatment for metastatic breast cancer and has intolerance to, or documented disease progression on or after their most recent therapy.
- * Has been previously treated with an anthracycline and/or taxane in the (neo)adjuvant or metastatic setting unless there was a medical contraindication to this treatment regimen.
- * Have lactate dehydrogenase (LDH) $\leq 2.5 \times$ upper limit of normal (ULN)

Part 2 Cohort C: Anti-PD-1/PD-L1 TrT-Naïve Solid Tumors-Liver:

- * Has histologically or cytologically confirmed Stage IV solid tumor that is not surgically resectable.
- * Complete resolution of toxic effect(s) of the most recent prior chemotherapy to Grade 1 or baseline (except alopecia). If participant received major surgery or radiation therapy of ≥ 30 Gray (Gy), they must have recovered from the toxicity and/or complications from the intervention.

4.2 Exclusion Criteria

All Arms and Cohorts (Parts 1 and 2):

- * Has had chemotherapy, definitive radiation, or biological cancer therapy within 4 weeks prior to the first dose of study drug, or has not recovered to Baseline or Common Terminology Criteria for Adverse Events (CTCAE) Grade 1 from the adverse events due to cancer therapeutics administered ≥ 4 weeks earlier.
- * Is currently participating and receiving study therapy or has participated in a study of an investigational agent and has received study therapy or has used an investigational device within 28 days of administration of ulevostinag. Note: Prior exposure to immunotherapeutics is allowed, including PD-1 and PD-L1 inhibitors
- * Is expected to require any other form of antineoplastic therapy while on study.
- * Is on chronic systemic steroid therapy in excess of replacement doses (prednisone ≤ 10 mg/day is acceptable), or on any other form of immunosuppressive medication.
- * Has a history of a second malignancy, unless potentially curative treatment has been completed, with no evidence of malignancy for 2 years.
- * Has clinically active central nervous system metastases and/or carcinomatous meningitis.
- * Has had a severe hypersensitivity reaction to treatment with a monoclonal antibody.
- * Has an active autoimmune disease that has required systemic treatment in the past 2 years.
- * Has a history of vasculitis.
- * Has an active infection requiring therapy.
- * Has a history of (non-infectious) pneumonitis that required steroids or current pneumonitis.
- * Has undergone prior allogeneic hematopoietic stem cell transplantation within the last 5 years.
- * Has Hepatitis B or C infection(s).

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- * Is pregnant or breastfeeding, or expecting to conceive or father children within the projected duration of the study.
- * Has not fully recovered from any effects of major surgery, and is free of significant detectable infection.
- * Has received a live vaccine within 30 days prior to first dose of study drug.
- * Has a history of re-irradiation for squamous cell carcinoma of the head & neck (HNSCC) at the projected injection site.
- * Has a tumor(s) in direct contact or encases a major blood vessel, and has ulceration and/or fungation onto the skin surface at the projected injection site.
- * HIV-infected participants with history of Kaposi's sarcoma and/or multicentric Castleman's disease
- * HIV-infected participants who have had an HIV-related opportunistic infection within 6 months
- * Has been treated with a Stimulator of Interferon Genes (STING) agonist (e.g. ulevostinag, ADU-S100).

All Part 2 Expansion Cohorts:

- * Has experienced weight loss $\geq 10\%$ over 2 months prior to first dose of study treatment.
- * Has clinically relevant ascites at baseline (defined as requiring paracentesis) or with moderate radiographic ascites. A minimal amount of radiographic ascites is allowed.
- * Has a history of interstitial lung disease.
- * For Part 2 Cohort C: Anti-PD-1/PD-L1 TrT-Naïve Solid Tumors-Liver, participants with MSI-H CRC are excluded.

11. Study Identifiers & Governance

Primary ID: 1048756961226174809

Registry Number: NCT03010176

Sponsor: Merck

Study Title: Study of Ulevostinag (MK-1454) Alone or in Combination With Pembrolizumab (MK-3475) in Participants With Advanced/Metastatic Solid Tumors or Lymphomas (MK-1454-001)

15. Sign-off & Approval

Principal Investigator: _____ Date: _____

Clinical Operations Lead: _____ Date: _____

Lead Statistician: _____ Date: _____